



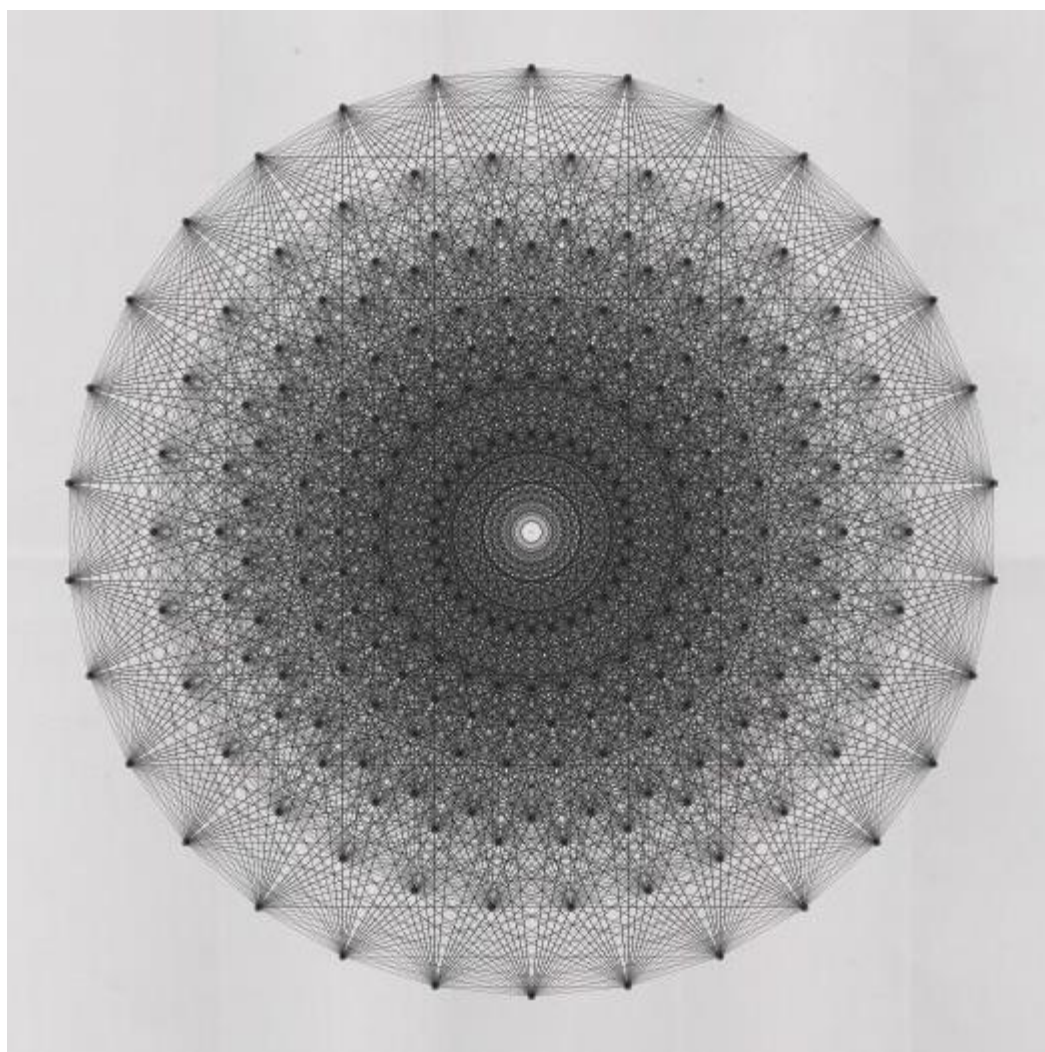
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COLLEGE



Tour de Maths 2017

Leg 5

Question Paper



5 Mark Questions

Question 1

Find the missing value.

$$\begin{array}{l} 1 + 4 \rightarrow 5 \\ 2 + 5 \rightarrow 12 \\ 3 + 6 \rightarrow 21 \\ 8 + 11 \rightarrow \square \end{array}$$

Question 2

Find the missing power for base 2.

$$2^{2017} - 2^{2016} = 2^{\square}$$

Question 3

It takes 10 minutes to fry one steak – 5 minutes from each side.

Two steaks can fit in your pan.

What is the shortest possible time to fry 3 steaks? Provide your answer in minutes.



Question 4

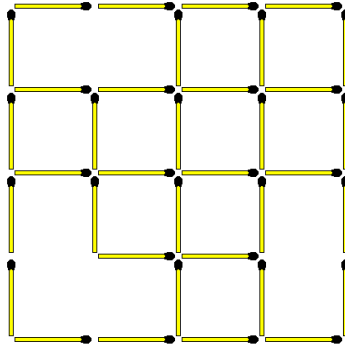
How many ways are there to divide 3 identical sweets between 3 friends?

(Note: all sweets must be distributed, not everyone has to get the same amount of sweets, and people can receive 0 sweets)



Question 5

How many complete squares are in this matchstick puzzle?

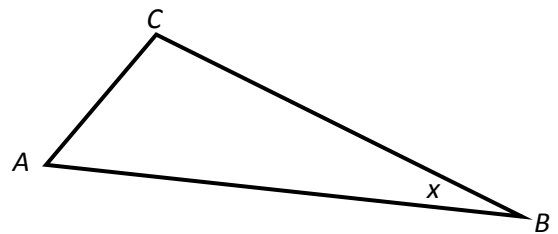


Question 6

Jess, Nicole and Caitlyn are three sisters. The product of their ages is 175. Jess and Nicole are twins. How old is Caitlyn?

Question 7

In triangle ABC, the angle at A is three times greater than the angle at B and the angle at C is double the angle at A. What is the value of the angle at A in degrees?



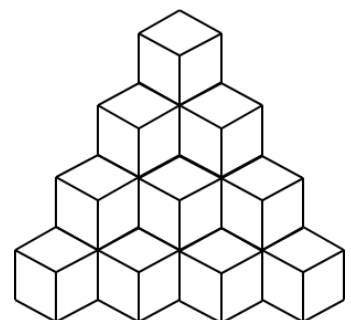
Question 8

$$\sqrt{x+15} + \sqrt{x} = 15$$

Find the smallest integer value for x.

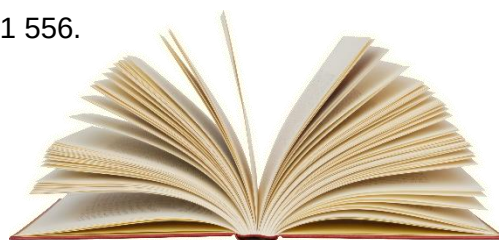
Question 9

The following stack of blocks that is 4 cubes high is made up of 20 cubes. How many cubes are required to build a similar stack that is 6 cubes high?



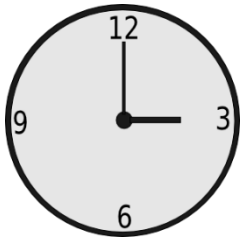
Question 10

The product of two consecutive pages of a book is 11 556. What is the page number of the page on the left?



10 Mark Questions

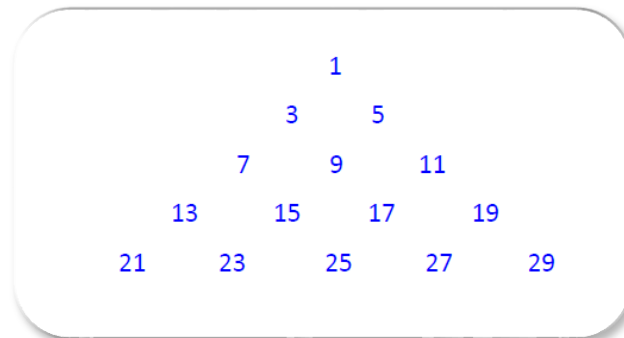
Question 11



Being an avid watch watcher, you discover that the hands of a watch are perpendicular at 3pm and at 9am.

How many times a 24-hour day are the hands of a watch perpendicular to each other?

Question 12



Find the sum of the 30th row.

Question 13

This question paper consists of 20 questions: 10 questions for 5 marks, 5 questions for 10 marks and 5 questions for 20 marks each. A team of 4 members, Brad, Muhammad, Conner and Ben get all the 5 mark questions correct and 4 of the 10 mark questions correct.

What is the minimum amount of 20 mark questions they must answer correctly to score over 80% for the paper as a whole?



Question 14

Determine the value of:

$$100^2 - 99^2 + 98^2 - 97^2 + \dots - 3^2 + 2^2 - 1^2$$

Question 15

If x denotes any negative integer number, which of the following expressions will always have the greatest value?

(A) $\sqrt{-x} \times \sqrt{-x}$

(B) $-(x^2 + 81)$

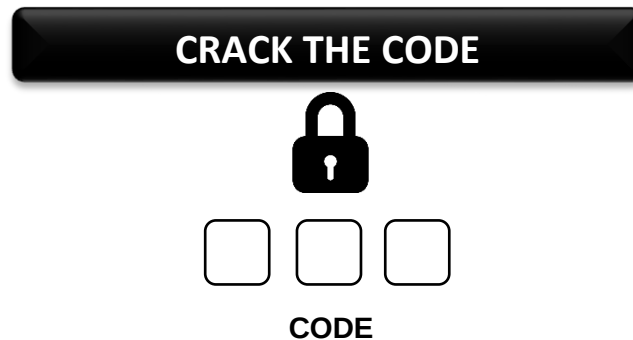
(C) $(\sqrt{x})^8$

(D) $\sqrt{x^4} + 1$

(E) $(\sqrt[3]{x^9})^3$

20 Mark Questions

Question 16



A numeric combination lock has a 3 digit key

682

One number is correct and well placed

614

One number is correct but incorrectly placed

206

Two numbers are correct but incorrectly placed

738

Nothing is correct

780

One number is correct but incorrectly placed

Question 17

There are four buttons in a row as shown below. Two of them show happy faces, and two of them show sad faces. If we press a face, its expression turns to the opposite (e.g. a happy face turns into a sad face after the touch). In addition to this, the adjacent buttons also change their expressions.

What is the least number of times you need to press a button in order to get all happy faces?



Question 18

$$\begin{array}{r} A \ B \\ + \ C \ D \\ \hline E \ F \end{array}$$

In this cryptogram, each letter represents a distinct (different) digit from 0 – 9. Zero may be used in the 'tens' column, e.g. AB could stand for 01.

What is the minimum possible value of the 2-digit integer EF?

Question 19

We define a relatively prime date to be a day for which the number of the month and the number of the day have no common factors other than 1. For example 22/5 (22 May) is such a day because 22 and 5 have no common factors other than 1.

Determine the month with the smallest number of relatively prime days.

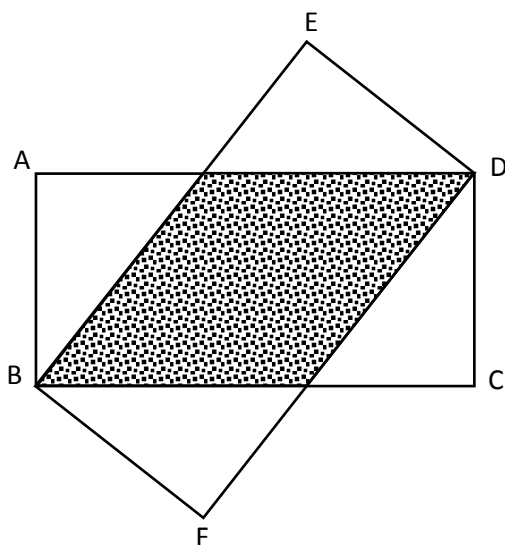


Question 20

In the diagram below, two identical rectangles have been placed on top of each other.

$AB = BF = 2$ and $AD = BE = 6$

Find the area of the shaded region.



Total: 200

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